

The Spartan Dawn Test: Breathing Hypersphere versus the Static Cosmos against CMB, BAO, and Supernovae

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Abstract

We confront the hyperspherical breathing-enclosure hypothesis with real cosmological data and let it fight for its life. The data are the Planck 2018 acoustic scale $100\theta_* = 1.04109$, twelve DESI DR2 baryon-acoustic-oscillation distances over $0.295 \leq z \leq 2.330$, and a Pantheon⁺ supernova shape prior $\Omega_m = 0.334 \pm 0.018$, with the sound horizon fixed at the first-principles value of Paper XI, $r_d = 147.09$ Mpc. Four models are fit by χ^2 minimization: flat Λ CDM, closed Λ CDM, flat $w_0 w_a$ CDM, and closed breathing $w_0 w_a$ CDM. The static closed hypersphere is rejected: given curvature as the only added freedom, the fit drifts to $\Omega_K = +0.004$ (faintly open) and the Akaike criterion does not justify the parameter ($\Delta\text{AIC} = +0.31$). The breathing models are strongly favoured ($\Delta\chi^2 = -8.6$ and -9.1), recovering an evolving equation of state $w_0 \approx -0.6$, $w_a \approx -1.4$ crossing $w = -1$ near $z \approx 0.5$ —the DESI evolving-dark-energy preference. The hypersphere is permitted but not forced: even the best closed fit pins $|\Omega_K| \lesssim 0.005$, $R_c \gtrsim 60$ Gpc. The honest verdict: the data do not kill the hot Big Bang—the sound-first ruler of Paper XI requires it—but they kill the static closed cosmos and wound the constant cosmological constant. Expansion-as-outward-motion is dethroned by duality: the atemporal breathing reading $H_0 = \dot{R}/R$ reproduces every distance. This is the empirical trial of ID10871, whose breathing term $F(a)$ now fits real data better than flat Λ CDM, leaving the derivation of $F(a)$ (Papers XVI–XVII) as the central open obligation.

1 Introduction: a child left in the wild

A hypothesis earns the right to be taken seriously by surviving contact with data it did not choose. The breathing-enclosure cosmology—in which the observed three-space is a closed S^3 whose radius $R(t)$ breathes, $H_0 = \dot{R}/R$ (ID10867)—makes claims the distance ladder can adjudicate. We fit it adversarially against flat Λ CDM and the naive static closed universe, using an information criterion that penalizes added parameters. One version dies, another thrives, and the hot Big Bang turns out to be required by the framework’s own first principle.

2 Two lenses on one observation

The temporal lens treats expansion as recession through time, $D_C(z) = c \int_0^z dz'/H(z')$. The atemporal hyperspherical lens treats redshift as a pure radius ratio $1 + z = R_0/R$ and distances as fixed chords on S^3 :

$$D_M(z) = R_0 \sin \chi(z), H_0 = \dot{R}_0/R_0.$$

Remark 2.1 (Duality). The two lenses are mathematically identical; no distance measurement distinguishes motion-through-time from geometric breathing. What data *can* measure is the sign of curvature, through the $\sin \chi$ (closed), χ (flat), or $\sinh \chi$ (open) radial map. The framework’s ontological claim costs nothing; the empirical question is whether the breathing is constant (Λ) or dynamic, and whether the enclosure curves closed.

3 Models and distance engine

With $E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_K(1+z)^2 + \Omega_X F(z)$ and

$$F(z) = (1+z)^{3(1+w_0+w_a)} e^{-3w_a z/(1+z)}, \quad w(a) = w_0 + w_a(1-a), \quad (1)$$

the curvature-aware distances are $D_M = (c/H_0 \sqrt{|\Omega_K|}) \sin(\sqrt{|\Omega_K|} H_0 D_C/c)$ for $\Omega_K < 0$, $D_H = c/H(z)$, $D_V = (z D_M^2 D_H)^{1/3}$. The four models are nested: flat Λ CDM, closed Λ CDM, flat $w_0 w_a$ CDM, closed breathing.

4 Data and fit

Fifteen data points: twelve DESI DR2 BAO distances (D_M/r_d , D_H/r_d , D_V/r_d across BGS, LRG, ELG, QSO, Ly α), the Planck acoustic scale $100\theta_* = 1.04109 \pm 0.00030$, and the Pantheon⁺ prior $\Omega_m = 0.334 \pm 0.018$; $r_d = 147.09$ Mpc fixed. Minimization is by Nelder–Mead; AIC = $\chi^2 + 2k$.

Table 1: Four-model fit to fifteen real data points.

Model	χ^2	k	Δ AIC	Best fit
A flat Λ CDM	16.13	2	0.00	$H_0 = 68.8$, $\Omega_m = 0.298$
B closed Λ CDM	14.44	3	+0.31	$\Omega_K = +0.004$
C flat $w_0 w_a$ CDM	7.51	4	−4.62	$w_0 = -0.61$, $w_a = -1.36$
D closed breathing	7.05	5	−3.08	$\Omega_K = -0.005$, $w_0 = -0.56$, $w_a = -1.37$

5 What died, and what survived

Proposition 5.1 (Static curvature is rejected). *Given curvature as the only added freedom (Model B), the fit prefers $\Omega_K = +0.004$ —faintly open, the wrong sign for a closed hypersphere—and the Akaike criterion (+0.31) does not justify the parameter. Curvature alone cannot shift the acoustic scale, the BAO ruler, and the supernova distances together without breaking one.*

Proposition 5.2 (Breathing is favoured). *Allowing w to evolve (Models C, D) improves the fit by $\Delta\chi^2 \approx -9$ for two parameters, with $w(a)$ crossing -1 near $z \approx 0.5$. The recovered $w_0 \approx -0.6$, $w_a \approx -1.4$ coincides with DESI DR2, an external result not tuned to.*

Remark 5.3 (Hypersphere permitted, not forced). Model D attains the best raw χ^2 but pins $|\Omega_K| \approx 0.005$, $R_c \gtrsim 60$ Gpc. A 3-sphere this vast looks locally flat.

6 On killing the Big Bang

A careless reading takes the breathing hypersphere to abolish the hot Big Bang. The sound-first ruler of Paper XI requires a hot, dense, ringing photon–baryon plasma; without it there is no sound horizon and the percent-level success of Paper XI evaporates. Sound-before-light is the hot beginning’s confirmation, not its removal. What the test kills is the static closed universe and the constant cosmological constant; what it reinterprets, by duality, is expansion-as-outward-motion. The surviving picture: the beginning was sound, expansion is geometric breathing of a vast nearly-flat enclosure, and the dark-energy pressure evolves.

7 Demarcation

Established physics: the FLRW distance–redshift relations; the closed-geometry $\sin \chi$ map; the DESI DR2 BAO measurements; the Planck acoustic scale; the Pantheon⁺ constraint; the DESI evolving-dark-energy preference at $2.8\text{--}4.2\sigma$; flatness bounds $|\Omega_K| \lesssim 0.002$.

Framework synthesis: the identification of the FLRW scale factor with a physical S^3 radius; the reading of the favoured dynamical dark energy as the enclosure’s breathing pressure $F(a)$ (ID10871); and the duality claim permitting the atemporal ontology at no statistical cost.

Predictions: (i) the $\sin \chi$ signature in high- z angular-diameter distances as future BAO push $|\Omega_K|$ below 10^{-3} ; (ii) redshift drift $\dot{z} = (1+z)H_0 - H(z)$, distinct from flat Λ CDM, within ELT/SKA reach; (iii) a correlation between the H_0 tension and the Λ anomaly if both are readouts of one breathing radius.

8 Limits of the construction

This is a compressed-statistics test—BAO summary distances, one CMB acoustic anchor, a supernova shape prior—not a full CMB-likelihood chain. It ranks the models and rejects static curvature; it is not a Boltzmann-code joint fit. The favoured improvement is carried by $F(a)$, not by Ω_K , which the data hold near zero, so the curvature claim remains permitted-but-unconstrained. And $F(a)$ is here fitted, not derived: converting that accommodation into a prediction is the obligation discharged in Papers XVI–XVII.

9 Conclusion: the survivor and its sword

Left in the wild, the static closed hypersphere died and the breathing one returned at dawn, carrying DESI’s evolving-dark-energy result as its sword. The data kill the static cosmos and the constant Λ , and leave the breathing enclosure permitted, favoured in its dynamical form, and vast if curved at all. Paper XIII computes the geometry behind the survivor and extracts from it a measurement of the spatial dimension itself.

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